A Conditional Theory of Permission and Obligation

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Abstract

This article argues for a conditional analysis of permission and obligation. The idea is that something is obligated just in case were it not to happen, the relevant normative ideals would necessarily be violated. Dually, something is permitted just in case were it to happen, the relevant normative ideals would possibly be met. I argue that obligation and permission are not only context dependent but also what I call *condition dependent*: the set of worlds relevant to evaluating a deontic claim varies with the statement at issue, in the same way that the evaluation of a conditional varies with its antecedent. I show that inheritance—the principle that whenever a statement is obligated, its logical consequences are too—is equivalent to antecedent strengthening for conditionals—*if A, C* entails *if B, C* whenever *B* entails *A*—and that the theory is equivalent to a modified Kratzerian framework in which the accessibility relation is condition dependent.

7,801 words

1 Conditions on Permissions

According to the standard semantics of deontic logic (Hanson 1965:180, Lewis 1973:100, Kratzer 1981b), something is permitted just in case it occurs in at least one deontically ideal accessible world. I present a challenge to this view. I then show that an alternative view, based on the work Alan Ross Anderson, Stig Kanger, and Risto Hilpinen, provides a much more satisfactory solution to the problem, compared to the standard theory.

Section 2 presents the problem and shows

2 The Problem

Two trains approach an intersection in a dense forest at full speed.One is travelling north to south, the other east to west. Each train driver can either apply the brakes or continue through the intersection. If both continue they will crash into one another. Table 1 lists the rules governing the train network.

Network Configuration		Status
A goes	B goes	Forbidden
A goes	B stops	Permitted
A stops	B goes	Permitted
A stops	B stops	Permitted

Table 1: The deontic status of each configuration of the train network.

The train network comes with a central control system that answers questions from the drivers. Each driver asks the system two questions: "Am I permitted to go through the intersection? Am I permitted to stop?" The system consults the rules and replies "Yes" to every question. It tells each driver that they are permitted to continue through the intersection and that they are permitted to stop.

Did the system respond correctly? If you were designing such an automated expert system to tell drivers what is permitted, would you design it to behave like this?

Intuitively, the system is faulty. There is something very wrong with its replies. Each driver would likely interpret the message to mean that things will be fine if they continue through the intersection. There is a clear danger that both drivers continue in response to the system's replies, resulting in catastrophe. Instead of a plain "Yes," it would have been much better for the system to reply instead, for example, "Indeterminate," "Not sure," or "It depends on what the other train does."

The problem is that our most prominent theory of permission today predicts the system's answers to be perfectly true in this scenario. According to the standard semantics of deontic logic (Hanson 1965:180, Lewis 1973:100, Kratzer 1981b), something is obligated just in case it holds in every deontically ideal accessible world, and permitted just in case it holds in at least one. It is natural to take the deontically ideal accessible worlds in this case to be those with a permitted configuration of the network: the worlds where the trains do not crash into each other. Then there is indeed a deontically ideal world where train A goes through (namely, where train A goes through and B stops), and also one where train B does (namely, where B goes through and A stops). So the standard semantics predicts the system's answers to be perfectly correct—the wrong result.[1]

In response to this challenge facing the standard theory, the goal of this article is to revive an older approach to deontic logic. It was suggested by Leibniz, and developed by Kanger and Anderson in the 1950s, before standard semantics was developed by von Wright and Hanson in the 1960s. The idea is this.[2]

> Something is obligated just in case if it did not happen, there would be no case in which the relevant normative ideals are violated.
>
> Something is permitted just in case if it happened, there would be some case in which the relevant normative ideals are met.

In other words, if you didn't do what is obligated, necessarily, the rules would be broken, while if you did what is permitted, possibly, the rules would be met. On this theory, conditional reasoning is central to the notions of obligation and permission, so I will call it the *conditional theory*.

The conditional theory provides, I believe, a much more satisfying account of the train case compared to the standard theory. As we will see below, the conditional theory predicts "Train A is permitted to go" and "Train B is permitted to go" to be unassertable, since the truth value of each depends on what the other train does. The standard theory, in contrast, predicts both to be straightforwardly assertable. Thus the standard theory predicts the train system's replies to be perfectly correct, while the conditional theory explains why they are not.

The conditional theory, as stated above, is a schema rather than a fully-fledged theory in its own right. It leaves two things to be specified: what underlying theory of conditionals we adopt, and what it means for the relevant normative ideals to be met.

Each choice affects what predictions for permission and obligation the conditional theory ultimately gives. First, it matters which underlying theory of conditionals we plug

[1]One may think that we may assert neither "Train A is permitted to go" nor "Train B is permitted to go" since neither train has *explicit* permission to go. However, there should be nothing wrong with the system explicitly permitting what is truly permitted. Since the standard theory predicts these sentences to be true, we would expect it to be fine for the system to state them. Moreover, we may contrast this with a case where the trains' paths do not cross. In that case, each train is intuitively permitted to go (something the standard theory correctly predicts). But still, neither train has been given explicit permission to go, making it unlikely that this response helps in the train case above.

[2]I have formulated the conditional in the past with subjunctive mood. This typically implies that the antecedent is false, though I regard this as an implicature (see Ippolito 2003, Leahy 2011, 2018) rather than part of the conditional's literal meaning. Following Stalnaker (1975), I take the subjunctive to be the more neutral form, imposing fewer requirements than the indicative; for instance, that the antecedent be compatible with the common ground, something I do not assume for the conditionals to come.

in. I show that inheritance—the principle that permission and obligation are closed under logical consequence—corresponds to antecedent strengthening in conditionals—the principle that *if A, C* entails *if B, C* whenever B entails A. The train case provides a counterexample to inheritance for permission. Intuitively, it is permitted that train A goes and train B stops, and it is permitted that train B goes and train A stops. But from this we do not want to conclude that for each train, it is permitted to go. Our result shows that to capture this on the conditional theory, the underlying theory of conditionals must invalidate antecedent strengthening.

Second, it matters how we analyse what it means for the relevant normative ideals to be met. I argue for a comparative interpretation. When we evaluate whether something is permitted or obligated, we compare its presence with its absence. Something is obligated just in case if it did not happen, necessarily, things would be worse than if it did, and something is permitted just in case if it happened, possibly, things will be no worse than if it did not. This comparative interpretation brings deontic reasoning closer to the kind of reasoning we see in decision problems, in which one compares what happens if an agent performs an action with what happens if they don't.

While the conditional theory may seem worlds apart from the standard theory, I show that a single change to the standard theory renders it equivalent to the conditional theory I propose (given the above interpretation of what it means for the relevant normative ideals to be met). The change is that, when evaluating whether a statement A is permitted or obligated, we take the accessible worlds to be those that result from supposing A or supposing its negation. I call such worlds *A-accessible*. Thus instead of one accessibility relation for each context, we have one for each combination of a context and statement. This is not part of the standard theory, in which the accessibility relation is taken to be independent of the statement at issue. Let us call an accessibility relation *condition dependent* just in case it depends on the statement at issue. I prove that A is obligated (respectively, permitted) according to the conditional theory if and only if A is true in all (some) A-accessible worlds. In this way, the difference between the standard and conditional theories comes down to a single source: condition dependence.

Here is the plan. Section 3 develops the conditional theory and shows how it accounts for the train case. In Section 4, I consider a reply on behalf of the standard theory that attempts to account for the train case via context dependence. Section 5 discusses correspondences that arise on the conditional theory between conditionals and deontic modality. Section 6 analyses what it means for the normative ideals to be met. Section 7 shows that the conditional theory is equivalent to a modified Kratzerian framework in which the accessibility relation is condition dependent, and compares the account with Jackson (1985) and Cariani's (2013) theories of *ought*. Section 8 concludes.

3 The Conditional Theory

I wish to revive an old analysis of deontic concepts, going back to Leibniz, and Kanger and Anderson's work in the 1950s (Kanger 1957, Anderson 1956, 1958, 1967; for an overview, see McNamara and Van De Putte 2022:§3).

In the *Elements of Natural Law*, Leibniz (2001) defines what is permitted as "what is possible for a good person to do" and what is obligatory as "what is necessary for a good person to do." Hilpinen (2017:159–60) suggests interpreting Leibniz's thought here using conditionals: something is permitted just in case if someone does it, possibly, they are a good person, and something is obligatory just in case if they don't do it, necessarily, they are not a good person.

Later, Kanger (1957) considered analysing *ought* A as $N(Q \supset A)$, where "N is the notion of analytic necessity," Q "a constant stating what morality prescribes," and $\supset$ the material conditional.[3] Around the same time, Alan Ross Anderson proposed that "to say that p is obligatory is to say that failure of p leads to a state-of-affairs $\mathscr{P}$ which is 'bad' " (Anderson 1958:103). He later wrote that "it is *analytic* of the notion of obligation that if an obligation is not fulfilled, then something has gone wrong" (Anderson 1967:346–47).[4]

We can formalise these ideas in a theory-neutral way using conditional selection functions. Let f be a selection function taking a statement A and a world w (and perhaps other parameters, depending on the theory in question) and returning the set of worlds relevant to evaluating a conditional whose antecedent is A.[5] There are many constraints one could impose on the selection function. I assume two minimal constraints: *success* (A is true at every selected A-world) and *reachability* (there is no selected A-world from w only if A is

[3]Hilpinen (1982) offers a similar analysis, with a variably strict conditional (Stalnaker 1968, Lewis 1973) in place of a strict conditional.

[4]There are some for whom such a conditional analysis of permission and obligation may seem entirely natural: speakers of Japanese, Korean, and Burmese. These languages most commonly express permission and obligation using conditionals. Kaufmann (2017a) calls these *conditional evaluative constructions*. For discussion, see Akatsuka (1992), Wymann (1996), Nauze (2008), Knoob (2008), Narrog (2009), Kaufmann (2017a), and Chung (2019). In a similar vein, Williamson (2007:157, 297) suggests that A is metaphysically necessary just in case if it were false, a contradiction would obtain, and dually, A is metaphysically possible just in case if it were true, no contradiction would obtain. That is, where $\top$ is a tautology and $\bot$ a contradiction, Williamson proposes to analyse $\Diamond A$ as $A \mathrel{\Diamond\!\!\rightarrow} \top$ and $\Box A$ as $\neg A \mathrel{\Box\!\!\rightarrow} \bot$. Williamson's analysis of necessity falls out as a special case of the pattern above when we interpret the good as the truth. For metaphysical modality this seems apt: what is 'good,' metaphysically speaking, is just what is true.

[5]To illustrate, for Lewis, $f(A, w)$ is the set of most similar worlds to w where A is true. (This formulation of Lewis's semantics makes the limit assumption; see Kaufmann 2017b.) On interventionist semantics of conditionals, $f(A, w)$ contains the result of intervening at w to make A true (Galles and Pearl 1998, Halpern 2000, Hiddleston 2005, Schulz 2011, Briggs 2012, Santorio 2019, Kaufmann 2013). I am neutral on the kind of conditional intended here; for example, whether it should be taken to be indicative or subjunctive.

a contradiction).[6]

In terms of selection functions, we may state the conditional analysis of obligation like so, with permission its dual. For convenience, let us call a world *good* just in case the relevant normative demands are met there.[7]

A is obligated at w	just in case	no world in $f(\neg A, w)$ is good.
A is permitted at w	just in case	some world in $f(A, w)$ is good.

Thus on the conditional approach, asking "May I open the window?" amounts to something like asking "Would it be ok if I opened the window?"

We are accustomed to calling a world an *A-world* just in case A is true there. Analogously, let us call a world an $if_w\, A$-world just in case it is in $f(A, w)$; that is, one we consider when supposing A. I will omit the subscript specifying the world of evaluation when it is clear from context.

The conditional theory has some immediately attractive features. First, permission is the dual of obligation: something is obligated just in case its negation is not permitted.[8] Second, obligation implies permission, provided there is at least one good world in $f(A, w)$ or $f(\neg A, w)$; in other words, provided we are not in a catch-22, where the relevant normative demands are guaranteed to be violated whether or not A holds.[9] Third, from the previous two, it follows that obligation is consistent: a statement and its negation are never both obligatory.

3.1 The Train Case on the Conditional Theory

Here is how the conditional theory accounts for the train case. To assess whether train A is permitted to go on the conditional theory, we ask whether things would be good if train A went. Now, for all we know, train B might go and it might stop. If we find ourselves in a world where train B goes, then if train A went, things would not be good, while if find ourselves in a world where B stops, then if train A went, things would be fine. Thus on the conditional theory, whether train A is permitted to go depends on what train B does.

[6] For formal details see Definition 1 of the Appendix.

[7] Section 6 discusses this notion of good in detail, deriving it from a preference order over worlds, representing betterness according to the relevant deontic ideals. For now, I rely on a simple binary classification of worlds into "good" and "not good." For formal details, see the Technical Appendix.

[8] Ensuring that obligation is the dual of permission—something is permitted just in case its negation is not obligated—requires the further assumption the selected $\neg\neg A$-worlds are just the selected A-worlds: $f(A, w) = f(\neg\neg A, w)$ for every world w.

[9] On the interpretation of *good* I offer in section 6, this *no catch-22* condition will follow from the assumption that among the *if A* and *if* $\neg A$-worlds, we can find a deontically ideal world: $f(A, w) \cup f(\neg A, w)$ contains a world that is no worse than any world in the set. This itself will follow from the limit assumption for the preference order over worlds.

Since we don't know what train B will do, we don't know whether train A is permitted to go, and therefore cannot assert so (assuming one may only assert what one knows; see, for example, Williamson 2000, McKinnon 2015). Similarly, we cannot assert that train B is permitted to go.[10] Figure 1 illustrates this reasoning for a toy model with only four worlds.

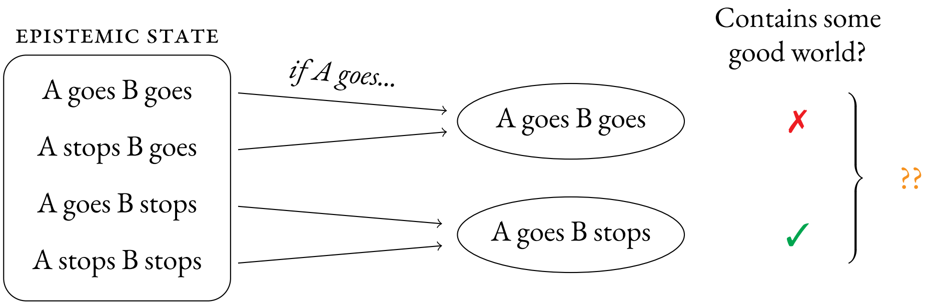

Figure 1: The conditional theory predicts "Train A is permitted to go" and "Train B is permitted to go" to both be unassertable.

This verdict of unassertability is, I believe, far more satisfying than the standard theory's prediction that "Train A is permitted to go" and "Train B is permitted to go" are both known to be true, and therefore assertable, in this scenario.[11] Figure 2 illustrates how the standard theory arrives at this result, again for a toy model with only four worlds.

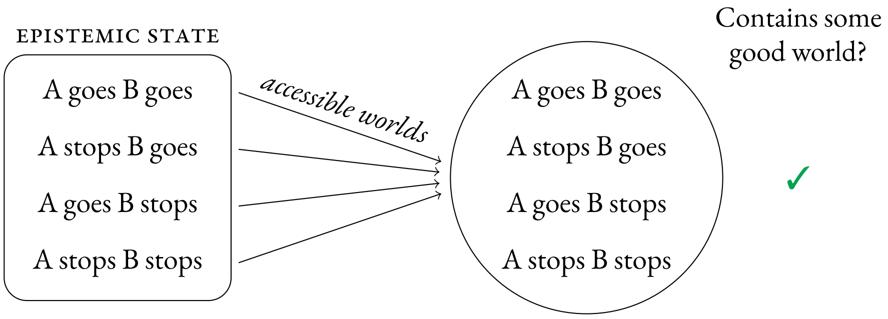

Figure 2: The standard theory predicts "Train A is permitted to go" and "Train B is permitted to go" to both be assertable.

[10] This also explains our discomfort in asserting the *negation*, that train A is not permitted to go. For all we know, train B might stop. In that case, A is permitted to go.

[11] Likewise, the conditional theory predicts "Train A must stop" and "Train B must stop" to be unassertable here, whereas the standard theory predicts them to be false. I have focused on permission since the contrast between unassertability and truth is, I believe, starker than the contrast between unassertability and falsity.

4 Context and Condition Dependence

This section considers a reply to the train case on behalf of the standard theory, appealing to the familiar context dependence of modals. As we see below, the train case calls for a theory in which the set of worlds used to evaluate deontic modals is not only context dependent, but also what I call *condition dependent*, sensitive not only to the context but also to the statement whose deontic status is at issue.

4.1 A Reply from the Standard Theory: Context Dependence

It is well known that deontic modals, like all modals, are notoriously context dependent. The early pioneers of deontic logic were well aware of this. Ernest Mally (1926) distinguishes the obligatory from the unconditionally obligatory. Georg von Wright, in the paper that gave birth to deontic logic as we know it today, notes that "In the smoking compartment, e.g., not-smoking is permitted and also smoking. But in the non-smoking compartment, not-smoking is permitted and smoking forbidden" (von Wright 1951:6). He mentions that while, on his account, "deontic propositions have been treated as 'absolute.' They can, however, be made 'relative' in several ways" (von Wright 1951:13). Things are permitted, obligated, or forbidden only under certain conditions. You may board the train if you have a ticket. You have to walk the dog, in view of the fact that he hasn't been outside yet today. You are forbidden from jumping out of the aeroplane unless you are wearing your parachute.

Angelika Kratzer has provided an influential treatment of the context dependence of modal expressions—here I focus on her approach. Kratzer proposes that deontic modals are *circumstantial*: they are evaluated in view of the circumstances.[12] The set of worlds in which these circumstances hold make up the *modal base*. She notes that we may equivalently express the worlds in the modal base as the accessible worlds, using a familiar Kripke-style accessibility relation, so in what follows, I will adopt this terminology. For Kratzer, a statement is a deontic necessity at a world w just in case it holds at every deontically ideal accessible world from w, and is a deontic possibility at w just in case it holds at some such world. Given its widespread popularity, I call this the *standard theory*.[13]

To assess Kratzer's predictions for "Train A is permitted to go," the key question is whether a world where A goes and B stops is compatible with the circumstances; that is, accessible. If it is, Kratzer's semantics predicts the sentence to be true, given the ordering over worlds in the train scenario, where a world is deontically ideal just in case there is no crash there.

[12] See Kratzer (1977, 1978, 2012), and especially Kratzer (1981b:53).

[13] For a formalisation of the standard theory, see Definition 8 of the Appendix.

Kratzer warns that "the kind of facts we take into account for circumstantial modality are a rather slippery matter" (Kratzer 1981b:53). Despite the slipperiness, we can rule out some obvious choices right away.

First, one might argue the circumstances are subject to a principle of accommodation. Kratzer mentions such a principle à la Lewis (1979b): "If the utterance of an expression requires a complement of a certain kind to be correct, and the context just before the utterance does not provide it, then ceteris paribus and within certain limits, a complement of the required kind comes into existence" (Kratzer 1981b:61). Unfortunately, this will not help here. Accommodation is a charity principle. It helps to make initially unassertable utterances assertable. Here we face the reverse problem, trying to explain why utterances (namely, such as "Train A is permitted to go" and "Train B is permitted to go") that the standard theory predicts to be assertable are in fact *un*assertable.

Second, one might propose that the circumstances include what each train will in fact do. Thus, for example, if train A in fact goes through the intersection at world w, the only worlds w accesses are worlds where train A does so. The trouble with this reply is that it trivialises deontic reasoning. According to it, if we happen to find ourselves in a world where both trains will go, both trains are *obligated* to go—clearly the wrong result. As a general strategy, whereby each world only accesses worlds agreeing with it, this amounts to each world only accessing itself, resulting in modal collapse: what is permitted and obligated is just what actually happens. This is none other than Leibnizian optimism, that we live in the best of all possible worlds, famously satirised by Voltaire's *Candide*. We often do what is not obligated, and even what is not permitted, a fact that any theory of permission and obligation ought to capture.

Third, one might propose that the accessibility relation decides on our behalf which train may go and which must stop; for instance, by accessing worlds where train A goes through but none where train B does. This would then imply that train A is permitted to go but train B is obligated to stop—a reasonable decision for the network operator to make. The central problem with this reply is that the scenario itself is perfectly symmetric with respect to the two trains. None of the information provided privileges granting permission to one train over the other. There is, therefore, no basis for the accessibility relation to behave in this way, to decide which train may go and which must stop. The train network operator may decide which train to permit to go, but a contextually-supplied accessibility relation itself does not.

Finally, one might propose that there is no accessible world where train A goes, and no accessible world where train B goes. This preserves the symmetry of the scenario, avoiding both of the train network system's problematic answers at once: that train A is permitted to go, and that train B is permitted to go. But this strategy faces a number of challenges.

Here are three. (I.) It predicts that each train is obligated to stop. Given that there is no accessible world A goes or where B goes, the only accessible worlds are those where both trains stop, making this obligated, which is intuitively incorrect. (II.) It fails to predict the truth of conditionals such as "If one train stopped, the other would be permitted to go." This follows on Kratzer's assumption that *if*-clauses restrict the set of accessible worlds (Kratzer 1986).[14] (III.) It fails to predict the truth of "It is permitted that train A goes and train B stops" and "It is permitted that train A stops and train B goes."

The solution to all of these problems is to allow the accessibility relation to access a world where train A goes, and one where train B goes. The problem for the standard theory is that, given the preference order in the train case (worlds with no crash are preferred to worlds with a crash), it then predicts "Train A is permitted to go" and "Train B is permitted to go" to both be true—the wrong result.

4.2 Deontic Modals are Condition Dependent

Why do the conditional and standard theories differ in their predictions for the train case? I think we can point to one difference in particular, what I call *condition dependence*. On the conditional theory, evaluating whether A is permitted involves asking what things would be like if A were true. Different suppositions result in a different division of what we fix and what we vary. In the train scenario, intuitively, what each train does is independent of what the other does. Supposing "If train A went" leaves fixed what train B does while allowing what train A does to vary; supposing "If train B went" does the opposite, leaving fixed what train A does while allowing what train B does to vary. Thus, on the conditional theory, asking "Is train A permitted to go?" results in a different division of what we fix and what we vary compared to asking "Is train B permitted to go?" In general, on the conditional theory, the circumstances (the facts we hold fixed) depend not only on context but also on the statement whose deontic status is at issue: deontic modals are not only context dependent, but also *condition dependent*.

Condition dependence is not part of the standard theory. It assumes that the circumstances we use to evaluate deontic statements (encoded via the accessibility relation) are provided by the conversational background—the general utterance situation, without regard for the statement under consideration.[15] On the standard theory, deontic modals are

[14] This is known as the *restrictor view*. For discussion see von Fintel (1994), Schulz (2010), Kratzer (2012), Mandelkern (2018), and Ciardelli (2022).

[15] As Kratzer (1981b:42) puts it, "Modals are context-dependent expressions since their interpretation depends on a conversational background which usually has to be provided by the utterance situation." Of course, each utterance trivially changes the context; namely, by changing the context into one in which that utterance has been made. But nothing in the rules governing the train network suggests that what is permit-

context dependent but not *condition* dependent. This lack of condition dependence, I argue, is why the standard theory gives the wrong result in the train scenario. For when we evaluate what one train is permitted to do, we fix what the other train does. For example, we fix what train B does when we evaluate whether train A is permitted to go, but not when we evaluate whether train B is permitted to go. Such variability in what we fix depending on the statement at issue calls for a condition-dependent theory.

Holly Martin Smith was the first to observe the condition dependence of deontic modals. Writing about a scenario similar to the train scenario above, she wrote that "the notion of 'evaluable from world *i*' would have to be relativized to the act or state of affairs whose modal status is at issue" (Smith 1977:246). There she was concerned with Lewis's proposal that something is obligated at a world w just in case it holds at all deontically ideal worlds from the standpoint of w. She notes that "incorporating this into Lewis's scheme would represent a substantial change in his theory, and it remains to be seen whether or not it could be successfully worked out" (Smith 1977:246–247). The insight was later incorporated into Jackson's theory of *ought* (Jackson 1985:187). Though to my knowledge, Smith's suggestion has never before been incorporated into a theory of permission, something the conditional theory proposed here aims to do.

5 Antecedent Strengthening and Inheritance

On the standard theory, but not the conditional theory, permission and obligation are closed under logical consequence, a principle called *inheritance*: if A entails B and A is permitted (respectively, obligated), then B is permitted (obligated) too.[16] This seems compelling at first glance. If you are permitted to run, plausibly, you are thereby permitted to move. However, the train scenario provides a counterexample to inheritance for permission. We are happy to assert that train A going and B stopping is permitted and that train A stopping and train B going is permitted, but as discussed, we do not want to assert that train A going is permitted and that train B going is permitted.[17]

Turning to conditionals, antecedent strengthening says that whenever A^+ entails A,

ted depends on what anyone utters; rather, it depends purely on whether or not there is a crash.

[16]The term "inheritance" has previously been reserved for the principle that *ought* is closed under logical consequence. For discussion see Cariani (2013) and Blumberg and Hawthorne (2023).

[17]We have put the point in terms of assertability, but we may just as well express it in terms of plain truth at a world. Suppose we are in a world where train B will in fact go through the intersection. In that world, intuitively train A is forbidden from going: it is false to say that train A is permitted to go. But it is nonetheless true that train A going and B stopping is permitted. (Granted, that won't happen, given that train B will go through, but as long we aren't fatalist—the idea that what happens is obligatory—even things that don't come to pass can be permitted.)

if A, C entails *if A^+, C*. To take Goodman's (1947) classic counterexample, it may well be true that if I struck this match, it would light, but false that if that I dipped this match in water and struck it, it would light. Similarly, if we are in a world where train B in fact goes, then "If A went, both trains would go" is true (given that the trains act independently), but "If A went and B stopped, both trains would go" is not.

On the conditional theory, antecedent strengthening corresponds to inheritance: inheritance is valid on the conditional theory if and only if the selection function satisfies antecedent strengthening.[18] It therefore predicts that failures of antecedent strengthening give rise to failures of inheritance. This is quite a remarkable result. It's fair to say that 'standard' theories of conditionals (that is to say, variably strict theories) violate inheritance, while 'standard' theories of deontic modals (treating them as quantifiers over worlds) satisfy inheritance.[19] On the conditional theory, this divergence is no longer tenable: reject antecedent strengthening for conditionals, and you must reject inheritance for deontic modals too.[20]

6 Which Worlds are Good?

On the conditional theory, A is permitted just in case some *if A*-world is good, and obligatory just in case no *if* $\neg A$-world is good. Which worlds are good?

A first thought is that the good worlds are the deontically ideal worlds, the worlds that are no worse than any world. This, however, is far too idealistic. Candide does not live in the best of all possible worlds. He, like us, lives in a world of earthquakes, tsunamis, fires, and wars. No matter what he does, he has no hope of living in a deontically ideal world out of all worlds whatsoever. That being said, Candide is still permitted to cultivate his

[18] We show this in Proposition 4 of the Appendix. A connection between antecedent strengthening and inheritance for *ought* has previously been observed by Jackson and Pargetter (1986:240).

[19] By 'standard theories of conditionals,' I have in mind Stalnaker (1968), Lewis (1973), and Kratzer (1981a, 1986). There has recently been renewed interest in strict conditional theories that validate antecedent strengthening and account for apparent counterexamples via context shifts; see Starr (2022) and the references therein. Jackson (1985), Lassiter (2011), and Cariani (2013) provide theories of *ought* that violate inheritance; for recent discussion of its status, see Blumberg and Hawthorne (2023).

[20] The failure of inheritance implies that, on the conditional theory, deontic logic is not a normal modal logic: normal modal logics validate the inference from $\Diamond(A \wedge B)$ to $\Diamond A$. Whether the conditional theory gives rise to a weaker modal logic depends one's underlying the theory of conditionals. Take, for instance, Substitution: the principle that *if A, C* and *if B, C* are logically equivalent (true in all the same worlds) whenever A and B are. Some theories of conditionals validate Substitution while others do not. Plugging one that does not into the conditional theory invalidates the analogous rule for permission and obligation: two sentences may be logically equivalent while only one is permitted or obligated. In that case, deontic logic would not even adhere to the popular neighbourhood semantics for deontic logic (Pacheco 2007, Pacuit 2017, Schlechta 2018).

garden. But even if he did, things would still not be deontically ideal. The earthquakes, tsunamis, fires, and wars would remain.[21]

A fix: shift from absolute to comparative good. What is the standard of comparison? A first thought is to look to the current circumstances. Something is permitted just in case if it occurred, things would be no worse than they actually are.

This solves the Candide problem: if Candide cultivates his garden, things will be no worse than they actually are. But it brings another. Even something that makes things better might not be good enough to reach the standards required for outright permission. Better often isn't good enough. Driving 10 km/h over the speed limit is better than driving 20 km/h over it (according to the traffic laws, we may suppose), but the traffic laws do not permit one who happens to be driving 20 km/h over the limit to drive 10 km/h over it, even though that would be less of a violation of the traffic laws. For Greek officials, if the British Museum repatriated half of the Parthenon sculptures and kept the rest, things would be better than they currently are (by Greek lights). Still, according to the Greek officials, the British Museum is not permitted to keep *any* of the sculptures: they maintain that repatriating half of the sculptures and keeping the rest is simply not something the museum is permitted to do. If a violation is not permitted, actually committing a bigger violation does not thereby permit the smaller one. We shouldn't be able to nullify a rule simply by committing a grosser violation of it.

Instead, we may compare a thing's occurrence with its absence.[22] Something is permitted just in case if it occurred, things would be no worse than if it didn't, some *if* A-world is no worse than any *if* $\neg A$-world, and obligatory just in case if it did not occur, things would be worse than if it did.

I adopt this comparative interpretation for the following reasons. First, it solves the Candide problem. Candide is permitted to cultivate his garden since doing so is no worse than not doing so. Second, it solves the better-but-not-good-enough problem. Driving 10 km/h over the speed limit is still not permitted, even for one driving 20 km/h over it. For if they drove 10 km/h over the limit, they are still not in an ideal world, since things could be better if they didn't; for instance, if they drove at or below the speed limit.[23]

[21] A version of this objection is raised by Nute (1985:177).

[22] Relatedly, Lewis (1973:100) considers the idea that *ought* A means A is better than its negation. For discussion and criticism, see Lassiter (2017:213).

[23] I pause to note that this reasoning also requires dropping conjunctive sufficiency: the principle that conditionals with true antecedents and true consequents are themselves true. In a world where A is false, when we suppose that it is false, conjunctive sufficiency would limit us to only considering that single world. Comparing *if* A to *if* $\neg A$ would collapse to comparing *if* A with the actual world, and big violations would render small violations permissible. Some theories of conditionals that invalidate conjunctive sufficiency include Fine's (2012) truthmaker semantics of counterfactuals and McHugh's (2022) aboutness theory.

Plugging this analysis of the good worlds into the conditional theory returns the following, which I take to be the official version of the conditional theory ($\leq$ denotes the preference order over worlds, where we define the strict version as usual: $u < v$ just in case $u \leq v$ but not $v \leq u$).

The Conditional Theory.

A is permitted at w	iff	some *if A*-world is no worse than any *if* $\neg A$-world
		$\exists w' \in f(A, w)\ \neg\exists w'' \in f(\neg A, w),\ w'' < w'$
A is obligated at w	iff	every *if* $\neg A$-world is worse than some *if A*-world
		$\forall w' \in f(\neg A, w)\ \exists w'' \in f(A, w),\ w'' < w'$

Let us say that a world is *good* with respect to statement A and world w (denoted $good_{A,w}$) just in case it is among the best *if A*- or *if* $\neg A$-worlds at w. Assuming that the preference order satisfies the limit assumption, we may equivalently state the conditional theory as follows. Let $\Diamond A$ denote that A is permitted, $\Box A$ that A is obligated, and let $A \Diamond\!\!\rightarrow C$ (respectively, $A \Box\!\!\rightarrow C$) be true at a world w just in case C holds at some (respectively, every) *if A*-world at w. We then have the following equivalences (Proposition 6 of the Appendix).

$\Diamond A$ is true at w	if and only if	$A \Diamond\!\!\rightarrow good_{A,w}$ is true at w.
$\Box A$ is true at w	if and only if	$\neg A \Box\!\!\rightarrow \neg good_{A,w}$ is true at w.

This theory gives the right results in all cases considered so far. It gives intuitive predictions in the train case, solves the Candide problem, and doesn't let big violations permit small violations.

7 Comparisons

7.1 The Standard and Conditional Theories

At first glance, the conditional theory may seem worlds apart from the standard theory. On the standard theory, A is permitted just in case it is true at some deontically ideal accessible world, and obligated just in case it is true at all of them.

The Standard Theory.

A is permitted	just in case	A is true at some ideal accessible world.
A is obligated	just in case	A is true at every ideal accessible world.

We may, nonetheless, trace the difference between the conditional and standard theories to a single source: a single switch that outputs the standard theory when set one way, and the conditional theory when set another way. The switch is to take the accessibility to be condition dependent, sensitive not only to the world of evaluation but also the statement at issue. Specifically, take the accessible worlds to be those that result from supposing the statement or its negation. For any statement A, let us say that a world v is *A-accessible* from a world w just in case v is a selected A-world at w or a selected $\neg A$-world at w. Thus instead of one accessibility relation, we have a family of accessibility relations, one for each statement.

$$wR_A v \quad \text{if and only if} \quad v \in f(A, w) \cup f(\neg A, w)$$

Given the limit assumption for the preference order over worlds, replacing accessibility with A-accessibility in the standard theory returns exactly the conditional theory (this is Proposition 9 of the Appendix).[24]

> A is permitted according to the conditional theory
> if and only if A is true at some ideal A-accessible world.
>
> A is obligated according to the conditional theory
> if and only if A is true at every ideal A-accessible world.

This equivalence shows that the difference between the standard and conditional theory all comes down to condition dependence. Plug in a condition-independent accessibility relation and you get the standard theory. Plug in the above condition-dependent accessibility relation and you get the conditional theory.

Our result above relating the standard and conditional theories also gives us a clear view of why Inheritance holds on the standard theory but fails on the conditional theory. Given that A^+ entails A, it follows that whenever A^+ is true at some (every) ideal accessible world, then so is A. But given that A^+ is true at some (every) ideal A^+-accessible world, it need not follow that A is true at some (every) ideal *A-accessible* world. For the A^+-accessible worlds and the A-accessible worlds will not, in general, be the same.

7.2 Jackson's Actualism

The conditional theory proposed here has much in common with Jackson's actualist semantics for *ought* (Jackson 1985, Jackson and Pargetter 1986). Jackson proposes that A is obligated out of alternatives $\{A, A^1, \ldots\}$ just in case the selected A-world is better than the selected A^i-world, for all i (1985:185).

[24] I take the limit assumption to state that every non-empty set of worlds contains a deontically ideal world (for details see Definition 2 of the Appendix). For discussion of the limit assumption, see Kaufmann (2017b).

Two differences stand out. First, the conditional theory proposed here always compares A with its negation, while Jackson regards this merely as a defeasible norm (Jackson 1985:188), allowing for other choices of alternatives. Second, the conditional theory allows the selection function to return multiple worlds, whereas Jackson formulates his theory assuming that it always outputs a single world, à la Stalnaker (1968). Jackson himself regards this as a "dubious assumption," but does not generalise his theory to overcome it. Assuming it, and that the alternative set consists of the statement and its negation, the conditional theory of obligation reduces to Jackson's.

Here I focus on the second difference. Jackson takes his theory to only apply to cases where we may reasonably assume that the selection function outputs a single world. Thus his theory is not designed to capture obligation for disjunctive acts like *take Metaphysics or Ethics*, or determinable acts like *paint the wall blue* (where many shades of blue are available), which the conditional theory proposed here aims to capture.

Do we really need this additional expressive power of the conditional theory over Jackson's? Here is an argument that we do. Jackson's theory is incompatible with the *coarseness* of obligation, the idea that something can be obligated even though there are impermissible ways for it to obtain (Wedgwood 2006:45, Cariani 2013:534). To illustrate, suppose Suzy works in a factory painting cars. Her instructions tell her to paint the upcoming car dark blue. Intuitively, Suzy is obligated to paint the car blue.[25] Here obligation is coarse. Suzy is obligated to paint the car blue, but some ways to do so—painting it light blue—are impermissible.

On Jackson's theory, Suzy ought to paint the car blue iff for every alternative to painting it blue, the selected world where she paints it blue is better than the selected world where she does the alternative. Plausibly, if Suzy paints it blue, the selection function could pick a world where she paints it light blue and it could pick one where she paints it dark blue.[26] Since it is indeterminate which selection function is at play, on Jackson's theory it

[25] Granted, asserting "Suzy ought to paint the car blue" is worryingly underspecified, and may be rejected on that basis. I take this rejection to arise from its pragmatic infelicity rather than falsity. Were it literally false, we would expect "Suzy is not obligated to paint the car blue" to be true. In fact it seems false, intuitively implying that Suzy is permitted to paint the car a colour other than blue. An additional confound here is the possibility of so-called metalinguistic negation (Horn 1985, Carston 1996), as in "Suzy is not obligated to paint the car BLUE. She is obligated to paint it DARK blue," featuring metalinguistic negation's telltale prosodic contour (see Lee 2017). Metalinguistic negation can target any objectionable feature of a sentence whatsoever (including pragmatic infelicity), illustrated by Horn's (1985) "I didn't trap two monGEESE, I trapped two monGOOSES."

[26] One might imagine that Suzy, being a diligent worker, always follows her instructions, so that if she paints it blue, she will certainly paint it dark blue. To counteract this, we may imagine that Suzy is reckless (she neglected to read her instructions, does not care about her duties, or the like), so that her painting it light blue is as plausible as her painting it dark blue. Even then, she is still obligated to paint the car blue.

is indeterminate whether she is obligated to paint the car blue.

In contrast, the conditional theory proposed here respects the coarseness intuition that it is determinately true that Suzy is obligated to paint the car blue. This is because every selected world where Suzy doesn't paint the car blue is worse than *some* selected world where she paints it blue (namely, one where she paints it dark blue). For those who take obligation to be coarse, this is a reason to prefer the conditional theory over Jackson's.

7.3 Cariani's Resolution Semantics

Cariani (2013) develops what he calls a resolution semantics for *ought*, framed in terms of a preference order $\succ$, a partition specifying the relevant ways of performing each action, and a benchmark: a ranked option "below which options are impermissible." He suggests that an action a is *permissible* just in case "some option that is a way of performing a meets or exceeds the benchmark," and *strongly* permissible just in case every such option is (Cariani 2013:8). Despite his theory's advantages, its predictions for the train case suffer from the same drawbacks as the standard theory. The most plausible partition for the train case specifies for each driver whether they go or stop. (The ordering could be, say, train A stops $\succ$ benchmark $\succ$ train A goes.) It is easy to see that no order over these options predicts the intuitive result that whether each train is permitted to go depends on what the other train does.

One might instead consider more refined options, specifying the actions of both drivers (such as both trains stop, A goes and B stops, B goes and A stops $\succ$ benchmark $\succ$ both trains go). This, however, violates Cariani's idea that the options represent the "ways" for agents to perform actions. For instance, train A going and B stopping is intuitively not a "way" for train A to go. There is nothing in Cariani's semantics to suggest how a sentence such as "Train A is permitted to go" is to be evaluated relative to this more complex set of options, involving the actions of multiple agents.

8 Conclusion

The conditional theory gives intuitive predictions in the train case. Where the standard theory predicts both "Train A is permitted to go through" and "Train B is permitted to go through" to be straightforwardly assertable, the conditional theory predicts them to be unassertable, since the truth of each depends on what the other train does. This shows that deontic modals are not only context dependent, but also condition dependent, sensitive to the statement whose deontic status is at issue. The theory overcomes the Candide problem, predicting that Candide is permitted to cultivate his garden, even though he has no hope of living in one of the best of all possible worlds. And the conditional theory does not allow a rule to be overridden simply by committing a bigger violation of it.

While I have sought to develop the conditional theory into a predictive account, many questions remain. First, it is naturally desirable to have a general theory of modality that captures what is common among the diverse modal flavours, such as epistemic, deontic, natural, alethic, metaphysical, and so on. It remains to be seen whether the conditional theory can be developed in a general framework for other modal flavours. Second, there has been great attention to the behaviour of logical words (such as disjunction and *any*) in permission statements, such as Ross's (1941) puzzle of free choice permission. This is especially exciting in light of the parallels between the behaviour of disjunction conditional antecedents and permission statements; for on the conditional theory, we might paraphrase "You may have coffee or tea" as "If you have coffee or tea, possibly, things will be good." Third, one may develop a dynamic conditional theory to shed new light on the problem of deontic update (Lewis 1979a, Yablo 2011).

Technical Appendix

Let $\mathcal{L}$ be the set of sentences of our language, closed (at least) under negation: for any $A \in \mathcal{L}$, $\neg A \in \mathcal{L}$ too.

Definition 1 (Model of the Conditional Theory)**.** A model is a tuple $(W, V, f, \leq)$ where

- W is a non-empty set;
- $V : \mathcal{L} \times W \to \{0, 1\}$ is a valuation, with $V(\neg A, w) = 1$ iff $V(A, w) = 0$ for $A \in \mathcal{L}$;
- $f : \mathcal{L} \times W \to \mathcal{P}(W)$ is a function satisfying, for every $A \in \mathcal{L}$,
  - *Success:* A is true at every selected A-world. $w' \in f(A, w)$ implies $V(A, w') = 1$ for all $w, w' \in W$;
  - *Reachability:* there is no selected A-world only if A is contradictory. For all $w \in W$, $f(A, w) = \emptyset$ implies $V(A, w') = 0$ for all $w' \in W$;
- $\leq \subseteq W \times W$ is a reflexive and transitive order.

Let $M = (W, V, f, \leq)$ be a model of the conditional theory, $w \in W$, and $A, B \in \mathcal{L}$. We define $A \mathrel{\Box\!\!\to} B$ (respectively, $A \mathrel{\Diamond\!\!\to} B$) to be true at M, w just in case $V(B, w') = 1$ for all (some) $w' \in f(A, w)$. As usual, we define that $u < v$ just in case $u \leq v$ but not $v \leq u$.

Definition 2 (The Limit Assumption)**.** We say a relation $\leq \subseteq W \times W$ *satisfies the limit assumption* just in case $<$ is well-founded: for any nonempty $S \subseteq W$, there is a $w \in S$ such that for no $w' \in S$ is $w' < w$.

Definition 3 (The Conditional Theory)**.** For any model $M = (W, V, f, \leq)$ of the conditional theory and world $w \in W$,

- A is permitted at M, w just in case for some $w' \in f(A, w)$, there is no $w'' \in f(\neg A, w)$ such that $w'' < w'$.
- A is obligated at M, w just in case for all $w' \in f(\neg A, w)$, there is no $w'' \in f(A, w)$ such that $w'' < w'$.

We define as usual that A *entails* B in a model M just in case $V(A, w) = 1$ implies $V(B, w) = 1$ for all $w \in W$. We say that *inheritance is valid on the conditional theory* just in case for any statements A and B, model M of the conditional theory and world w, if A entails B, then if A is obligated at w, so is B, and if A is permitted at w, so is B.

Note that, given duality, inheritance for permission and inheritance for obligation are equivalent. For instance, if A entails B, by contraposition, $\neg B$ entails $\neg A$, so by inheritance for permission, $\Diamond\neg B$ entails $\Diamond\neg A$, so $\neg\Diamond\neg A$ entails $\neg\Diamond\neg B$. Then by duality, $\Box A$ entails $\Box B$, which gives us inheritance for obligation.

We define that a selection function f *satisfies antecedent strengthening* just in case for any A and B such that A is not a contradiction and B is not a tautology, if A entails B then $f(A, w) \subseteq f(B, w)$ for any world w. (A statement A is a contradiction in a model M with valuation V just in case $V(A, w) = 0$ for all $w \in W$, and a tautology in M just in case $V(A, w) = 1$ for all $w \in W$.) We only consider cases where A is not a contradiction and B not a tautology to make room for hyperintensional theories of conditionals (such as Fine 2012) that distinguish between logically equivalent antecedents. For on some theories of conditionals, A may be contradictory (and hence entail any B) without $f(A, w)$ being a subset of $f(B, w)$ for any B whatsoever; or B may be a tautology (and hence be entailed by any A), without $f(A, w)$ being a subset of $f(B, w)$ for any A whatsoever.

Proposition 4. *Inheritance is valid on the conditional theory if and only if the selection function satisfies antecedent strengthening.*

Proof. We prove this for permission. Obligation is similar (and also follows from duality).

($\Leftarrow$) Suppose the selection function f satisfies antecedent strengthening. Pick any model $M = (W, V, f, \leq)$ of the conditional theory with that selection function f. Suppose that A entails B and that A is permitted at M, w. Then for some $w' \in f(A, w)$, $w'' < w'$ for no $w'' \in f(\neg A, w)$. By success, A is true at w', so A is not a contradiction. If B is a tautology, B is vacuously permitted at M, w (since W is non-empty, B is true at some world, so by reachability, $f(B, w)$ is non-empty, and by success, $f(\neg B, w)$ is empty).

So suppose B is not a tautology. Then by antecedent strengthening, $f(A, w) \subseteq f(B, w)$. So $w' \in f(B, w)$. Suppose towards a contradiction that $w'' < w'$ for some $w'' \in f(\neg B, w)$.

Since A entails B, by contraposition, $\neg B$ entails $\neg A$. Since A is not a contradiction, $\neg A$ is not a tautology, and since B is not a tautology, $\neg B$ is not a contradiction. Then by antecedent strengthening, $f(\neg B, w) \subseteq f(\neg A, w)$. Hence $w'' \in f(\neg A, w)$, contradicting the fact that A is permitted at w. B is therefore permitted at w.

($\Rightarrow$) Suppose the selection function f does not satisfy antecedent strengthening. Then there are some statements A and B such that A entails B, B is not a tautology, and for some world w, $f(A, w) \nsubseteq f(B, w)$. Then there is some $w' \in f(A, w)$ such that $w' \notin f(B, w)$. Consider the model $M = (W, V, f, \leq)$ where the preference order $\leq$ satisfies

$$w' < \text{all worlds in } f(\neg B, w) < \text{all worlds in } f(B, w) < \text{all other worlds.}$$

(Precisely, 'all other worlds' denotes $W \setminus (\{w'\} \cup f(\neg B, w) \cup f(B, w))$.) This preference order is well defined since the sets are disjoint: $w' \notin f(B, w)$, and since $w' \in f(A, w)$, by success, A is true at w', then as A entails B, B is true at w', so $w' \notin f(\neg B, w)$; also by success, $f(B, w)$ and $f(\neg B, w)$ are disjoint. Then A is permitted at M, w, since $w' \in f(A, w)$ and, by construction, w' is better than every world whatsoever, so there is no $w'' \in f(\neg A, w)$ such that $w'' < w'$.

Suppose towards a contradiction that B is permitted at M, w. Then for some $v \in f(B, w)$, there is no $v' \in f(\neg B, w)$ such that $v' < v$. Since every world in $f(\neg B, w)$ is preferred to every world in $f(B, w)$, $f(\neg B, w)$ must be empty. Then by reachability, $\neg B$ is a contradiction, so B is a tautology, contradicting the fact that it is not. $\square$

Definition 5 (The good worlds)**.** Given a sentence A and world w, we define that a world $w' \in W$ is $good_{A,w}$ in a model of the conditional theory just in case it is in $f(A, w) \cup f(\neg A, w)$, and for no $w'' \in f(A, w) \cup f(\neg A, w)$ is $w'' < w'$.

Proposition 6. *Let $M = (W, V, f, \leq)$ be a model of the conditional theory such that $\leq$ satisfies the limit assumption. Then for any sentence A and world w,*

- *A is permitted at M, w according to the conditional theory iff $A \Diamond\!\!\rightarrow good_{A,w}$ is true at M, w;*
- *A is obligated at M, w according to the conditional theory iff $\neg A \Box\!\!\rightarrow \neg good_{A,w}$ is true at M, w.*

Proof. We prove the permission case. The obligation case is similar.

Suppose A is permitted at M, w. Then for some $w' \in f(A, w)$, there is no $w'' \in f(\neg A, w)$ such that $w'' < w'$. Consider $S = \{w'' \in f(A, w) \cup f(\neg A, w) : w'' < w'\}$. If S is empty, w' is good, and since $w' \in f(A, w)$, $A \Diamond\!\!\rightarrow good_{A,w}$ is true at w. So suppose S is nonempty. Then by the limit assumption, there is a $w^* \in S$ such that $w'' < w^*$ for no $w'' \in S$. Then w^* is $good_{A,w}$, and $w^* \in f(A, w) \cup f(\neg A, w)$. Since $w^* < w'$, $w^* \notin f(\neg A, w)$. So $w^* \in f(A, w)$. Hence $A \Diamond\!\!\rightarrow good_{A,w}$ is true at M, w.

Conversely, suppose $A \diamondto good_{A,w}$ is true at M, w. Then for some $w' \in (A, w)$, $w'' < w'$ for no $w'' \in f(A, w) \cup f(\neg A, w)$. A fortiori, $w'' < w'$ for no $w'' \in f(\neg A, w)$, so A is permitted at M, w. □

Definition 7 (Model of the Standard Theory)**.** A model of the standard theory is a tuple $(W, V, R, \leq)$ where W, V, and $\leq$ are as before (Definition 1), and $R \subseteq W \times W$.

Given an order $\leq$ and set of worlds P, let $\textsc{best}_{\leq}(P)$ be the set of deontically best worlds among P: $\textsc{best}_{\leq}(P) = \{w \in P : \neg\exists w' \in P, w' < w\}$, and let $R[w] = \{w' \in W : wRw'\}$ denote the set of worlds accessible from w.

Definition 8 (The Standard Theory, Kratzer 1981b)**.** For any model $M = (W, V, R, \leq)$ of the standard theory and world $w \in W$,

- A is permitted at M, w just in case $V(A, w') = 1$ for some $w' \in \textsc{best}_{\leq}(R[w])$;
- A is obligatory at M, w just in case $V(A, w') = 1$ for all $w' \in \textsc{best}_{\leq}(R[w])$.

Proposition 9 (Equivalence with Condition-Dependent Kratzer)**.** *Let $M = (W, V, f, \leq)$ be a model of the conditional theory and w a world. Suppose that $\leq$ satisfies the limit assumption. For every sentence $A \in \mathcal{L}$, define a relation $R_A \subseteq W \times W$ such that wR_Av iff $v \in f(A, w) \cup f(\neg A, w)$ for any worlds $w, v \in W$.*

Then A is permitted (obligated) at M, w according to the conditional theory if and only if A is true at some (every) ideal A-accessible world at M, w: $V(A, w') = 1$ for some (every) $w' \in \textsc{best}_{\leq}(R_A[w])$.

Proof. Here we show the equivalence for permission. The obligation case is similar. Suppose A is permitted at w according to the conditional theory: for some $x \in f(A, w)$, $y < x$ for no $y \in f(\neg A, w)$. Consider $S = \{y \in f(A, w) : y \leq x\}$. Since $x \in S$, S is non-empty. Then by the limit assumption, there is a $y \in S$ such that $z < y$ for no $z \in S$. Then $y \in f(A, w)$, so by success, A is true at y.

We show that y is a deontically ideal A-accessible world. For suppose it's not. Then $z < y$ for some $z \in f(A, w) \cup f(\neg A, w)$. Since $y \in S$, $y \leq x$, and so $z < x$. If $z \in f(A, w)$, then since $z \leq x$, $z \in S$, contradicting the fact that $z < y$ for no $z \in S$. If instead $z \in f(\neg A, w)$, then as $z < x$, this contradicts the fact that $z < x$ for no $z \in f(\neg A, w)$.

Conversely, if A is true at some deontically ideal A-accessible world at w, then for some A-world $y \in f(A, w) \cup f(\neg A, w)$, $z < y$ for no $z \in f(A, w) \cup f(\neg A, w)$. A fortiori, $z < y$ for no $z \in f(\neg A, w)$. If y were in $f(\neg A, w)$, by success, $\neg A$ would be true at y, contradicting the fact that A is true at y. So y must be in $f(A, w)$. A is therefore permitted at w according to the conditional theory. □

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